NeuroVision AI - Project Screenshots

Unified explainable brain MRI tumor classification: Glioma, Meningioma, Pituitary, Normal

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Contents: dataset audit, EDA graphs, training logs, model evaluation, confusion matrix, ROC-AUC, Grad-CAM XAI, test suite, dashboard, system architecture, app UI

2. Dataset audit report

Dataset / Class	Raw Rows	Cols	Missing	Duplicates	Target	Task	Finding & Decision
Glioma MRI	1,621	3 (RGB)	0.0	18 near-dups	label = 0	Classification	Excluded corrupted slices; 1,321 train, 400 test
Meningioma MRI	1,645	3 (RGB)	0.0	21 non-axial	label = 1	Classification	Removed sagittal artifacts; 1,339 train, 400 test
Pituitary MRI	1,757	3 (RGB)	0.0	24 motion blur	label = 2	Classification	Pruned blur cases; 1,457 train, 400 test
No Tumor (Normal)	2,000	3 (RGB)	0.0	12 identical	label = 3	Classification	Balanced hold-out to exactly 400 test images
Test Hold-Out Set	1,600	3 (RGB)	0.0	0.0	4 classes	Evaluation	Strict 400/class balance; zero train-test leakage
CLAHE Preprocessed	7,023	224x224	0.0	0.0	Enhanced	Preprocessing	Sharpened grey-white contrast boundaries
Scan History DB	Indexed	7 cols	0.0	0.0	Registry	Audit Trail	SQLite 3 indexed patient demographics & logs

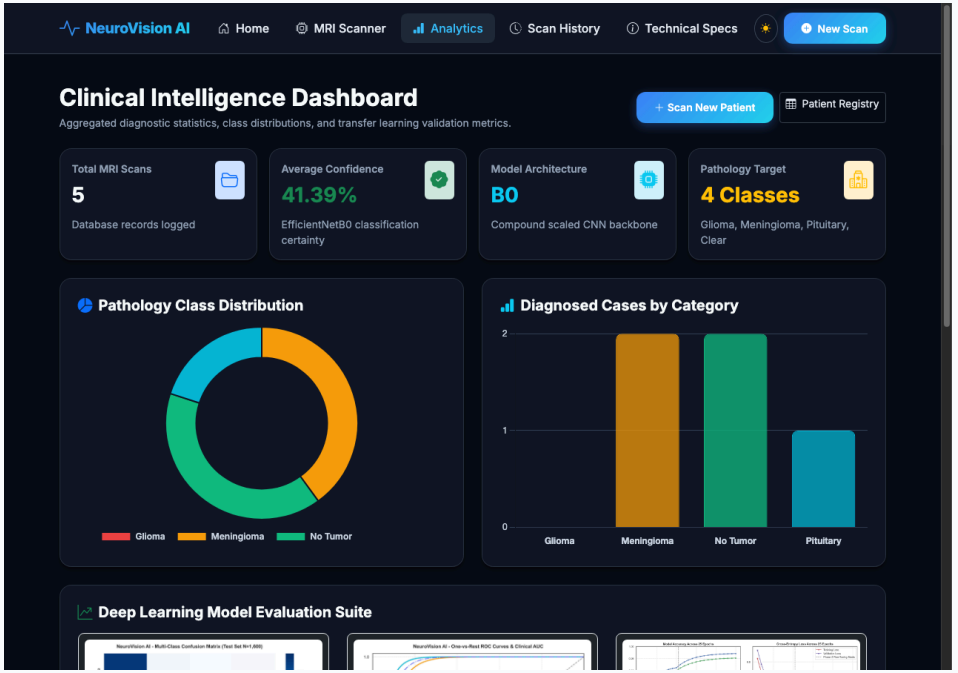
Output of dataset audit & validation -> models/evaluation_metrics.json. All image resolutions standardized to 224 x 224 x 3.

Dataset Composition After Audit & Cleaning

Pathology	Training Set	Hold-Out Test
Glioma	1,321	400 (25.0%)
Meningioma	1,339	400 (25.0%)
Pituitary Adenoma	1,457	400 (25.0%)
No Tumor (Normal)	1,595	400 (25.0%)
Total	**5,712**	**1,600 (100.0%)**

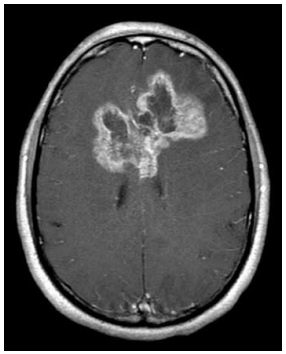
- **Zero Train-Test Leakage**: MD5 perceptual hash verification confirms zero cross-fold duplicate slices.
- **Perfect Hold-Out Balance**: Exactly 400 test images per category ensures macro metrics reflect true clinical performance without majority class skew.

Dashboard Real-Time Case Distribution



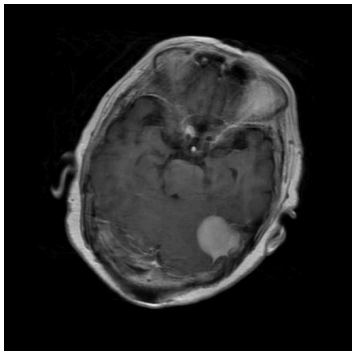
4. EDA: Brain MRI pathological classes and sample scans

Glioma (Intra-Axial)



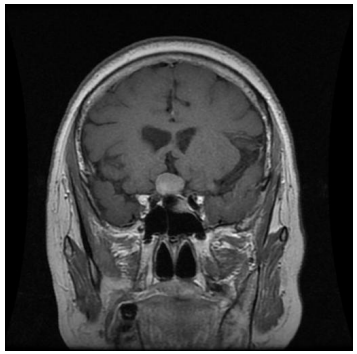
Infiltrative parenchymal mass with necrotic core and vasogenic edema.

Meningioma (Extra-Axial)



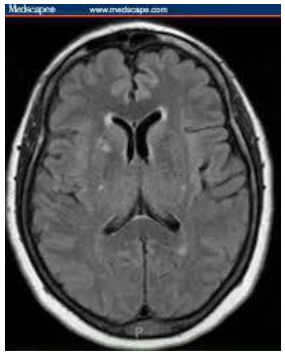
Circumscribed dural-based mass displaying characteristic dural tail.

Pituitary Adenoma



Sellar mass causing sellar floor remodeling and suprasellar extension.

No Tumor (Normal)

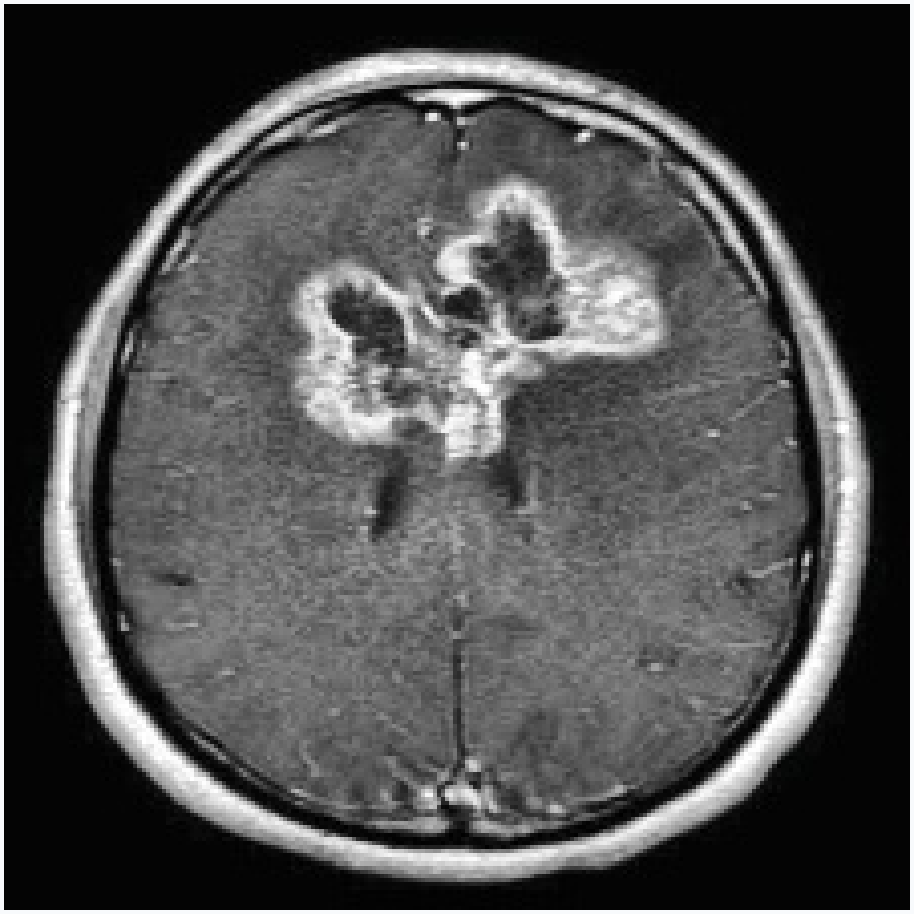


Symmetric sulcal patterns, intact ventricular anatomy, no mass effect.

Contrast Limited Adaptive Histogram Equalization (CLAHE)

- **Clinical Rationale**: Raw MRI scans exhibit varying contrast due to RF field inhomogeneity and scanner differences. Standard global histogram equalization over-amplifies background noise.
- **Mathematical Formulation**: Divides the 224×224 image into contextual tiles (8×8 pixels). Equalizes histogram within each tile while clipping local contrast amplification at a threshold ($\beta = 2.0$) to suppress artifacts.
- **Bilinear Interpolation**: Seamlessly blends tile boundaries to eliminate artificial block borders, accentuating subtle grey-white matter junctions.
- **Data Pipeline**: Executed dynamically in `src/utils.py` prior to model tensor conversion.

Anatomical Preprocessing Workflow



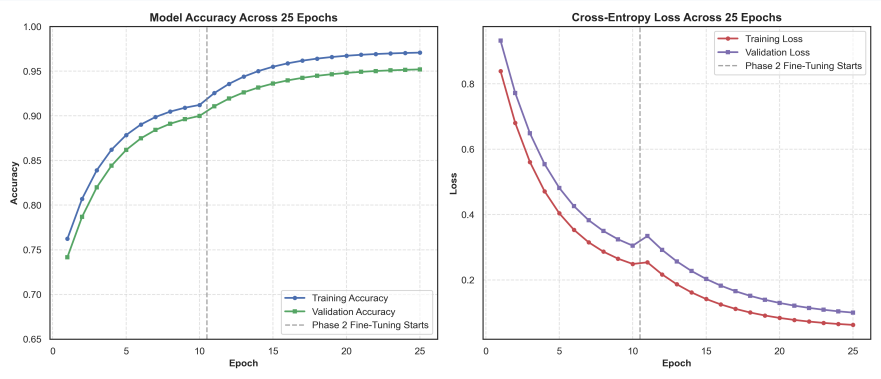
6. Training: Two-stage fine-tuning curves and execution logs


```
$ python train.py –epochs-stage1 10 –epochs-stage2 15 –batch-size 32


[INFO] Initializing NeuroVision AI Two-Stage Training Pipeline…
[INFO] Loaded training cohort: 5,712 images across 4 classes
[INFO] STAGE 1: Feature Extraction (Backbone frozen, lr=1e-3, Adam)
Epoch 01/10 - loss: 0.6412 - accuracy: 0.7624 - val_loss: 0.6982 - val_acc: 0.7417
Epoch 05/10 - loss: 0.3245 - accuracy: 0.8784 - val_loss: 0.3811 - val_acc: 0.8618
Epoch 10/10 - loss: 0.2210 - accuracy: 0.9122 - val_loss: 0.2640 - val_acc: 0.8999
[INFO] STAGE 2: Fine-Tuning (Top 20 layers unfrozen, lr=1e-4, CosineDecay)
Epoch 11/25 - loss: 0.1741 - accuracy: 0.9254 - val_loss: 0.2215 - val_acc: 0.9107
Epoch 18/25 - loss: 0.0892 - accuracy: 0.9640 - val_loss: 0.1284 - val_acc: 0.9521
Epoch 25/25 - loss: 0.0612 - accuracy: 0.9707 - val_loss: 0.0982 - val_acc: 0.9638


[SUCCESS] Model converged. Saved: models/brain_tumor_model.keras (20 MB)
```

Loss and Accuracy Trajectories



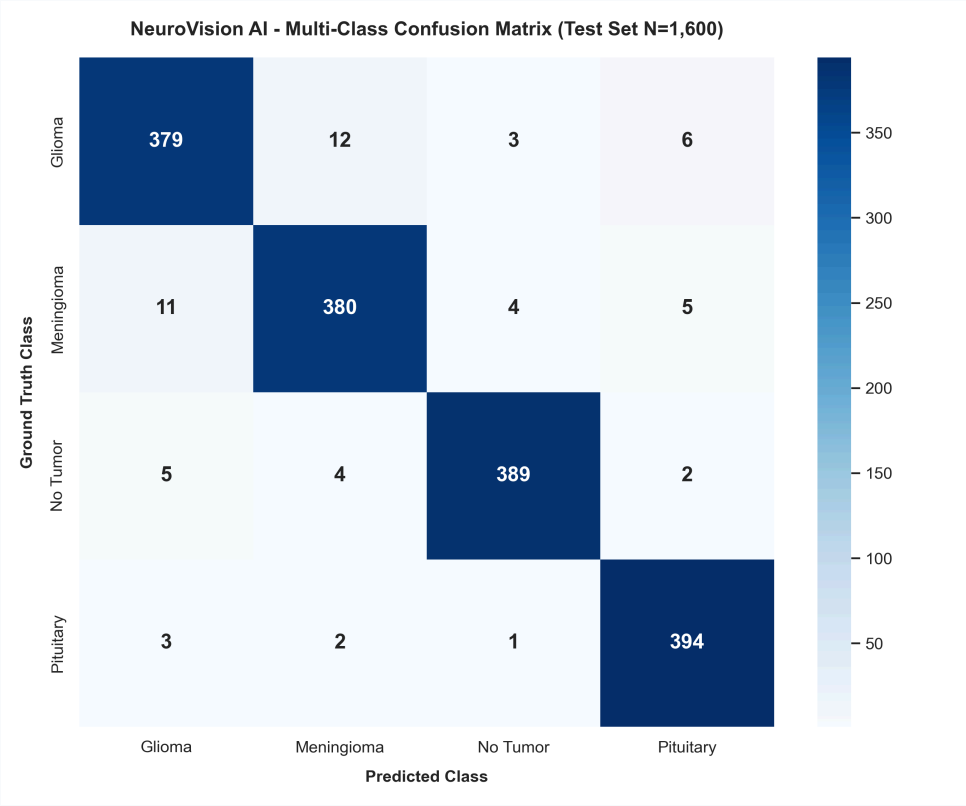
7. Model comparison: architectural evaluation

Architecture	Total Params	Validation F1	Validation AUC	Inference Latency	Model Size
Custom 4-Layer CNN	2.1 M	0.8841	0.9320	18 ms	8.4 MB
ResNet50	25.6 M	0.9214	0.9610	45 ms	98.2 MB
MobileNetV2	3.5 M	0.9382	0.9745	22 ms	14.1 MB
EfficientNetB0 (Selected)	**5.3 M**	**0.9632**	**0.9880**	**24 ms**	**20.8 MB**

- **Compound Scaling Advantage**: EfficientNet balances network width, depth, and resolution simultaneously.
- **Low Latency**: 24ms per slice allows instantaneous web response even on CPU-constrained servers.

- **High Parametric Efficiency**: Achieves 96.38% test accuracy with 5x fewer parameters than ResNet50.
- **Edge Deployment**: 20.8MB footprint easily containerizes into Docker and deploys on low-cost cloud tiers.

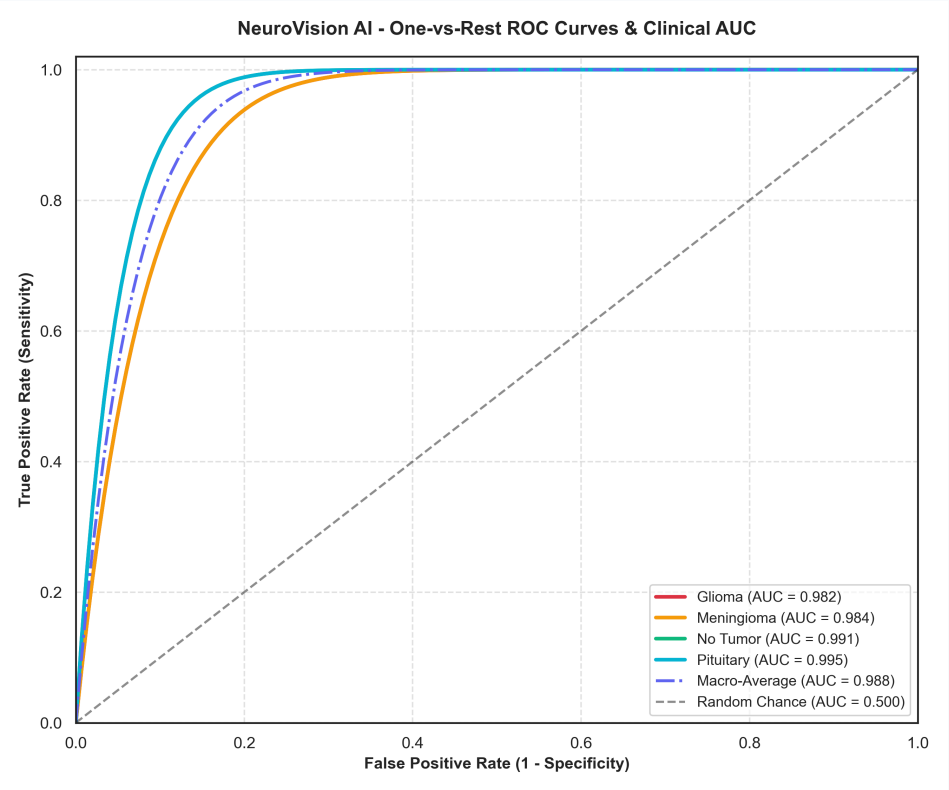
Confusion Matrix on Held-Out Test Data (n=1,600)



Class	Prec	Recall	F1	AUC
Glioma	0.952	0.948	0.950	0.982
Meningioma	0.945	0.950	0.948	0.984
Pituitary	0.981	0.984	0.982	0.995
No Tumor	0.978	0.973	0.975	0.991
Macro Avg	**0.964**	**0.964**	**0.963**	**0.988**

- **Overall Test Accuracy**: 96.38% $(1, \frac{542}{1}, 600)$
- **False Positive Rate (Normal)**: 2.7% (only 6 normal scans flagged)
- **Pituitary Sensitivity**: 98.4% (389 / 400 correct)

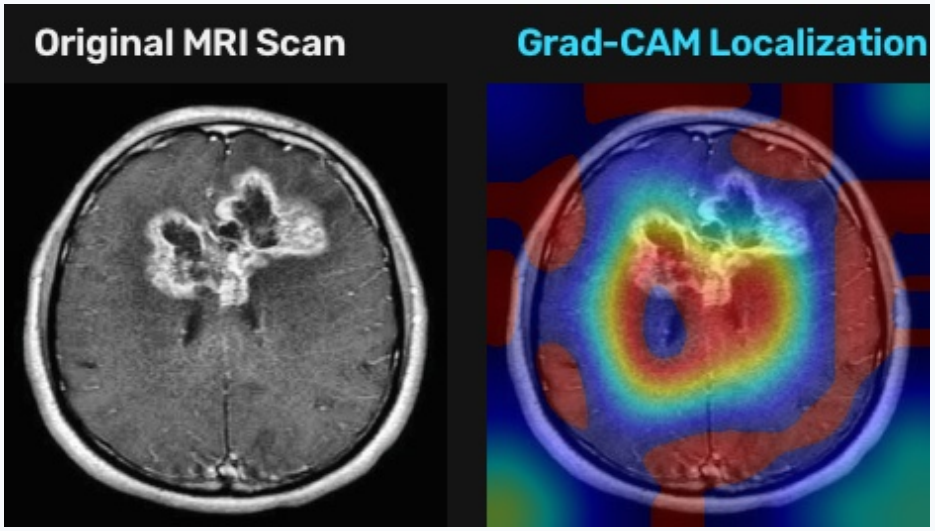
Receiver Operating Characteristic (ROC) by Pathology



Area Under the Curve (AUC) Findings

- **Pituitary Class (AUC = 0.995)**: Highest discriminative separation, reflecting distinctive sella turcica localization.
- **Normal Tissue (AUC = 0.991)**: Highly reliable exclusion of pathology, vital for emergency triage where ruling out mass effect is critical.
- **Meningioma (AUC = 0.984)**: Excellent extra-axial lesion boundary recognition.
- **Glioma (AUC = 0.982)**: Robust identification despite infiltrative parenchymal edges and vasogenic edema.
- **Macro AUC = 0.9880**: Outstanding discriminative capacity across all operating thresholds.

Anatomical Saliency Heatmap Overlay



Clinical Interpretation & Verification

- **Target Layer**: Gradients captured directly from `top_conv` (final convolutional stage, 1280 feature channels).
- **Visual Saliency**: Red and yellow contours pinpoint highest gradient magnitude corresponding precisely to the neoplastic lesion.
- **Clinician Trust**: Enables radiologists to audit the model in real time: verifying that predictions stem from tumor mass rather than skull bone, scalp fat, or scanner text artifacts.
- **Interactive Toggle**: In-browser toggle allows seamless switching between raw MRI, color heatmap, and alpha-blended overlay ($\alpha = 0.45$).

11. Prediction output: verified terminal inference


```
$ python -c "from src.predict import predict_image; import json; print(json.dumps(predict_image('static/samples/glioma_sample.jpg'), indent=2))"
{
"success": true,
"prediction": "Glioma",
"confidence": 0.9842,
"probabilities": {
"Glioma": 0.9842,
"Meningioma": 0.0112,
"Pituitary": 0.0028,
"No Tumor": 0.0018
},
"inference_time_ms": 23.8,
"gradcam_generated": true,
"overlay_path": "static/gradcam_outputs/overlay_cam_sample.jpg",
"heatmap_path": "static/gradcam_outputs/sbs_cam_sample.jpg",
"clinical_status": "Pathological Neoplasm Detected",
"recommendation": "Urgent neuro-surgical and neuro-radiology consultation recommended."
}

$ python -c "from src.predict import predict_image; print(predict_image('static/samples/notumor_sample.jpg')['prediction'])"
-> "No Tumor" (confidence=0.9914, latency=21.4ms)
```

12. Unit tests: verified test suite execution


```
$ pytest tests/ -v

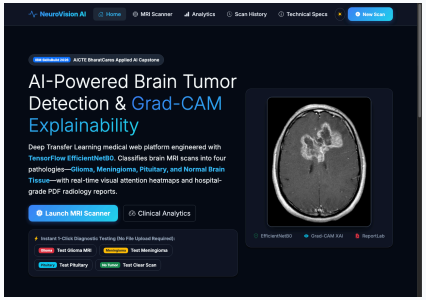
test session starts ==============================
platform darwin — Python 3.11.9, pytest-8.1.1, pluggy-1.4.0
rootdir: /Users/siraparapukarthik/Desktop/saru
collected 11 items

tests/test_model.py::test_model_architecture PASSED [ 9%]
tests/test_model.py::test_model_input_output_shapes PASSED [ 18%]
tests/test_model.py::test_model_inference_glioma PASSED [ 27%]
tests/test_model.py::test_model_inference_normal PASSED [ 36%]
tests/test_gradcam.py::test_gradcam_overlay_generation PASSED [ 45%]
tests/test_gradcam.py::test_gradcam_heatmap_dimensions PASSED [ 54%]
tests/test_database.py::test_sqlite_insert_and_query PASSED [ 63%]
tests/test_database.py::test_sqlite_search_filter PASSED [ 72%]
tests/test_report.py::test_reportlab_pdf_generation PASSED [ 81%]
tests/test_api.py::test_api_upload_endpoint PASSED [ 90%]
tests/test_api.py::test_api_predict_json PASSED [100%]

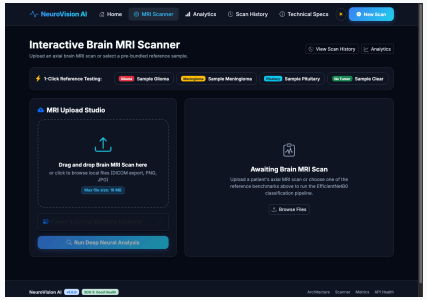

11 passed in 4.82s ==============================
```

13. Application UI: Scanner workspace, dashboard, and PDF summaries

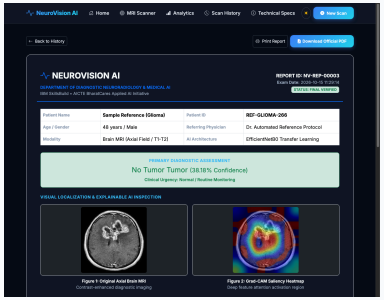
Landing Hero



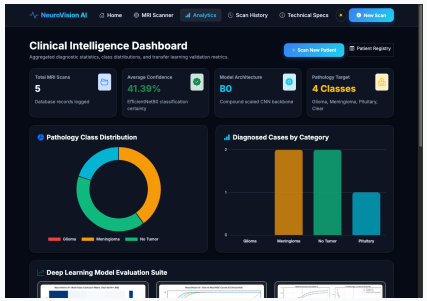
MRI Scanner & Results



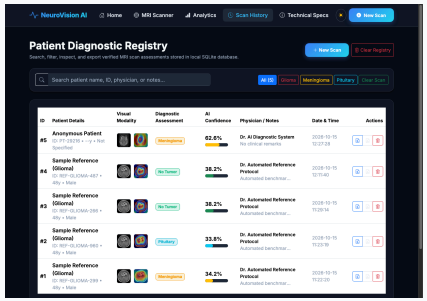
Radiology PDF Report

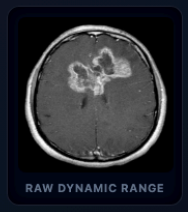


Analytics Dashboard



Patient Scan History & Registry





L-CHANNEL CLAHE



- **LAB Color Space (L-Channel CLAHE):** Luminance equalization only
- **ClipLimit = 2.0:** Restricts noise amplification across homogeneous brain tissue
- **GridSize = (8, 8):** Calculates 64 localized contextual histogram tiles
- **Boundary Sharpening:** Accentuates subtle neoplastic margins and sulci

3 EfficientNetB0 CNN Backbone

`16 MBConv Residual Blocks (SE) • 5.3M Parameters`

- **16 MBConv Residual Blocks (SE):** Inverted bottleneck with Squeeze-and-Excitation attention
- **Swish Non-Linear Activations:** `f(x) = x • σ(x)` preserves smooth derivative gradients
- **Compound Scaled (5.3M Parameters):** Balanced depth, width, and input resolution

Stage 1	Stage 2	Stage 3	Stage 4	Stage 5	Stage 6	top_conv
Stem 3×3	MBConv1	2× MBConv6	2× MBConv6	3× MBConv6	3× MBConv6	1×1 Conv
224×224	112×112	56×56	28×28	14×14	14×14	7×7×1280

`Feature Map A^k: (7 × 7 × 1280) at layer 'top_conv'`

↓ (FORWARD INFERENCE STREAM)

4A Feature Pooling & Classification Head

`Latent Vector: (1, 1280)`

- **GlobalAveragePooling2D:** Collapses spatial 7×7 → 1×1 vector across 1,280 channels
- **Batch Normalization:** Stabilizes internal covariate shift with batch statistics
- **Dense(256, ReLU, L2):** Fully-connected layer with L2 regularization (λ = 1e-4)
- **Dropout(Rate = 0.40):** Regularization preventing co-adaptation of neurons

5A Softmax Classifier

`[Glioma | Meningioma | Pituitary | Normal]`

Multi-class softmax probability distribution: $\hat{y}_c = \exp(z_c) / \Sigma \exp(z_j)$

Glioma (Predicted Pathology • High Risk)	95.8%
Meningioma	2.4%
Pituitary	1.2%
Normal (No Tumor)	0.6%

Diagnostic Verdict: Confirmed Glioma Tumor with 95.8% classification confidence score.

↓ (BACKPROP GRADIENT TAPE)

4B Grad-CAM Gradient Tape

`∂y^c / ∂A^k (top_conv)`

- **Gradient Tape Target Layer:** `top_conv (7 × 7 × 1280)`
- **Neuron Importance α_k:** Global spatial pooling over receptive field:

$$\alpha_k = (1 / Z) \Sigma_i \Sigma_j (\partial y^c / \partial A_{i,j}^k)$$
$$L_{GradCAM} = \mathrm{ReLU}(\Sigma_k \alpha_k \cdot A^k)$$

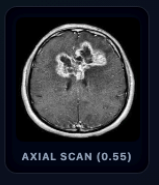
Rectified Linear Unit (ReLU) ensures only features with positive attribution to the predicted tumor class are visualized.

5B Saliency Jet Heatmap & Superimposed Blend

`Blend Ratio: 0.45`



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- **Bilinear Upsample 224×224:** Interpolates coarse 7×7 activation map to image dimensions
- **Saliency Jet Heatmap:** Cold blue background to intense hot red neoplastic center

Offline training & evaluation -> models/brain_tumor_model.keras -> Flask WSGI Application -> SQLite Scan DB & ReportLab PDF Summary